

Assignments and Arithmetic Operators

ELEC1006: ENGINEERING COMPUTING

Operators

- Assignment (=)
- Arithmetic operators (+, -, *, /, %)
- Compound assignment (+=, -=, *=, /=, %=, >>=, <<=, &=, ^=, |=)
- Increase and decrease (++ , --)
- Relational and equality operators (<, >, <=, >=, ==, !=)
- Logical operators (!, &&, ||)
- Bitwise operators (&, |, ^, ~, <<, >>)
- Conditional operator (?)
- Comma operator (,)
- Bitwise operator
- Explicit type casting operator
- Precedence of operators

Assignment (=)

- Assignment operator assigns a **value** to a **variable**

a=5;

- Most important rule in 'assignment' is 'right-to-left', i.e., the operation always takes place from the right to left. **a=b;** (whatever in **a** will be overwritten by the contents of **b**)
- **a=2+(b=5);**  **b=5; a=2+b;**
- **a=b=c=5;**  **c=5; b=c; a=b;**

Multiple Assignments

- The **=** can be used to assign a value to multiple variables:

$$x = y = z = 5;$$

- Value of **=** is the value that is assigned
- Associates right to left:

$$x = (y = (z = 5)) ;$$

Combined Assignment

- Look at the following statement:

New value of sum	Old value of sum
	
sum	sum

= sum + 1;

sum + 1 adds 1 to the variable sum and stores the result back into variable sum using the assignment =.

Assignments & Arithmetic Operators

Statement	What It Does	Value of x After the Statement
<code>x = x + 4;</code>	Adds 4 to x	10
<code>x = x - 3;</code>	Subtracts 3 from x	3
<code>x = x * 10;</code>	Multiplies x by 10	60
<code>x = x / 2;</code>	Divides x by 2	3
<code>x = x % 4</code>	Makes x the remainder of x / 4	2

Arithmetic Operators

(+, -, *, /, %)

- % is the modulo operation (**not a percentage calculation**) – it calculates the remainder of the division between the two values:

`a=11 % 3;`  `a = 2;`

Algebraic Expressions

Algebraic Expression	Operation	C++ Equivalent
$6B$	6 times B	<code>6 * B</code>
$(3)(12)$	3 times 12	<code>3 * 12</code>
$4xy$	4 times x times y	<code>4 * x * y</code>

Algebraic Expressions

- Multiplication requires an operator:
 $Area = lw$ is written as $Area = l * w;$
- There is no exponentiation operator:
 $Area = s^2$ is written as $Area = pow(s, 2);$
- Parentheses may be needed to maintain order of operations:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

is written as

$$m = (y_2 - y_1) / (x_2 - x_1);$$

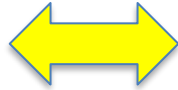
NOT

$$m = y_2 - \frac{y_1}{x_2} - x_1$$

$$m = y_2 - y_1 / x_2 - x_1;$$

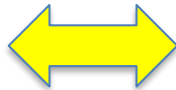
Compound Assignment

`a += 5;`



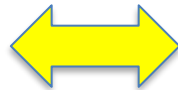
`a = a + 5;`

`a -= 5;`



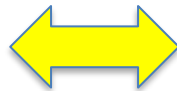
`a = a - 5;`

`a /= b;`



`a = a / b;`

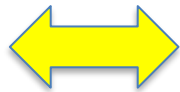
`a *= b+1;`



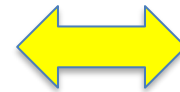
`a = a * (b+1);`

Increase and decrease

`a ++;`



`a += 1;`



`a = a + 1;`

Compound Assignment

Increase and decrease

<code>a ++;</code>		<code>a += 1;</code>		<code>a = a + 1;</code>
<code>++a ;</code>				<code>a = a + 1;</code>

If a=2, b=5

<code>c=a ++*b;</code>		<code>c=a*b;</code> <code>a=a++;</code>	<code>c=10;</code> <code>a=3;</code>
<code>c=++a *b;</code>		<code>a=a++;</code> <code>c=a*b;</code>	<code>a=3;</code> <code>c=15;</code>

Same rules apply to --

More info on C++

- [1] cplusplus.com: Operators
<https://cplusplus.com/doc/tutorial/operators/>